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# Static pure strain sensing using dual–comb spectroscopy with FBG sensors

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**Abstract:** We propose a method to precisely characterize the optical response of a fiber Bragg grating (FBG) sensor for static strain sensing by using dual–comb spectroscopy (DCS). By digitally post–correcting the mutual noise between the two combs, a robust and pure strain sensing system is achieved by compensating for the temperature–induced frequency shift of the FBG sensor. The comb–resolved radio–frequency (RF) spectra generated by DCS are obtained. Meanwhile, the stability of the central Bragg frequency of the FBG–reflected spectrum in the RF domain is 0.315 kHz, which is better than the difference of repetition rate (1 kHz). A reference FBG is used for detecting and compensating for the temperature–induced central frequency drift. Finally, a spectral sensitivity of 0.85 pm/με with 0.8 με static strain resolution is achieved.

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## 1. Introduction

Fiber Bragg gratings (FBGs) are wavelength–encoded sensors that have been successfully demonstrated for measurements of parameters such as strain, temperature, and vibration [1–3]. Compared with conventional electrical sensors, FBG sensors, with unique advantages of immunity to electromagnetic interference, cost–effectiveness, and compact design, provide the opportunity to be embedded in composite materials and concrete structures [4, 5]. In general, FBG strain sensors are evaluated by characteristics including sensitivity, resolution, dynamic range, and multiplexing capability. Specially, static FBG strain sensors with high resolution and high sensitivity are required in seismic and volcanic monitoring [6]. The static sensing here refers to the measurement for slow change of strain, which is typically below 10 Hz [7, 8]. Essentially, the Bragg wavelength of each FBG sensor is defined as $\lambda_B = 2n_{eff}\Lambda$, where $\lambda_B$ is the Bragg wavelength, $n_{eff}$ is the effective refractive index of the FBG and $\Lambda$ is the grating period. Therefore, the key–point for FBG applications is the demodulation of the variation of the central Bragg wavelength [9].

Commonly, FBG sensor for strain sensing is integrated with a broadband amplified spontaneous emission (ASE) source or a tunable continuous–wave laser. With an ASE source, the Bragg wavelength $\lambda_B$ can be demodulated in the frequency domain. An optical spectrum analyzer (OSA) is used to collect the FBG–reflected spectrum. However, the resolution of strain sensing is limited by the low spectral resolution of OSA (typically, 0.1 nm) as well as the long–time for measurement. Moreover, some ASE sources, such as light–emitting diodes, usually require a complex optical power compensation system for reducing intensity noise [10]. The tunable continuous–wave laser is another choice for strain sensing with high sensitivity, where the modulation of wavelength is time–consuming and the resolution is limited by laser frequency noise [3, 11]. Besides, the intrinsic noise of tunable continuous–wave lasers is much larger at low frequencies than at higher frequencies, making it more difficult to measure static strain (< 10 Hz) than dynamic strain [7].

Optical frequency combs (OFCs) offer hundreds of thousands of synchronized low–noise

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laser lines in a broad spectrum. Each mode can be expressed by the repetition frequency ($f_{rep}$) and carrier–envelope offset frequency ($f_{ceo}$) as $f(n) = nf_{rep} + f_{ceo}$ [12, 13]. To fully detect and utilize these comb lines, researchers initially proposed a dual-comb interferometer using two combs with slightly different repetition rates ($\Delta f_{rep}$), in which the optical comb lines are down–converted to the radio–frequency (RF) domain. Multi–heterodyne spectroscopy and an RF comb are generated [14, 15]. Recently, dual–comb interferometry has been recognized as a powerful technique that provides high–sensitivity broadband spectroscopy, ultra–high resolution and fast acquisition speed. Practical applications include gas analysis [16–18], absolute distance measurements [19–21], ellipsometry [22], material characterization [23–25], hyperspectral imaging [26], and vibrometry [27]. OFCs have already been used for precise strain measurements based on a change in repetition frequency [28]. The dual–comb spectroscopy (DCS) in FBG strain sensing would serve simultaneously as a broadband light source and a demodulation method. With intrinsic asynchronous sampling between the two pulse trains, the interference signals are easily obtained.

Previously, Kuse *et al.* presented a static FBG strain sensing system using the DCS technique, which enabled measurements of the FBG spectrum in the low–frequency radio domain with a photo–detector (PD) [7]. Nevertheless, the complicated tight–locking system between the two combs renders it inconvenient for in–situ applications. Additionally, the temperature–induced Bragg frequency shift may cause measurement error for static pure strains because no temperature compensation method was used in Kuse's sensing system. Guo *et al.* demonstrated multiplexed static FBG strain sensing using DCS with a free–running fiber laser [29]. Since the two frequency combs were generated by a free–running laser source, the mutual noise of the two pulse trains can be minimized. However, the frequency fluctuation of the peak reading of one of the FBGs with no strain is much larger than the tooth spacing in the RF spectrum, which prevents the determination of the central frequency of the FBGs within the adjacent longitudinal mode and limits the measurement precision of the central frequency of the FBG–reflected spectrum as well as the strain–induced central frequency shifts.

In this paper, we demonstrate a temperature–independent strain FBG sensing system based on dual–comb spectroscopy, in which the mutual noise is corrected using a digital processing method. Without tight–locking between the two combs, the light source system is greatly simplified. Benefitting from the post–correction method (see our previous publication for more details [30]), a comb–resolved and phase–stable dual–comb interferometer is developed. In our work, the error for the determination of the central frequency of the FBG is smaller than the repetition rate difference i.e. $\Delta f_{rep}$ in the RF domain without any tight–locking; therefore, the central frequency of the FBG–reflected spectrum can be distinguished within the adjacent comb teeth. We also show that this pure strain sensing system based on DCS enables high sensitivity and resolution in the static regime.

## 2. System setup and principle of operation

### 2.1. Experimental setup for static strain sensing

In this section, we describe the basic working principles of DCS with digital noise compensation and its application to pure static FBG strain sensing. A common dual–comb interferometer system consists of two frequency combs with slightly different repetition frequencies ($\Delta f_{rep}$). Since the repetition frequencies as well as the carrier–envelope offset frequencies of the two laser sources, are locked to the microwave reference, the frequency modes can be stabilized, which can serve as an ultra–precise frequency ruler in optical frequency metrology. By coupling the outputs of the two combs, the two pulse trains sample each other, with very fine effective time shifts between each consecutive pair of pulses given by $\Delta T = \Delta f_{rep}/f_{rep}(f_{rep} + \Delta f_{rep})$ over a time interval $1/f_{rep}$, as is shown in Fig. 1(a). This process is analogous to linear optical sampling or a sampling oscilloscope, in which the sampling gate given by the femtosecond pulses of one

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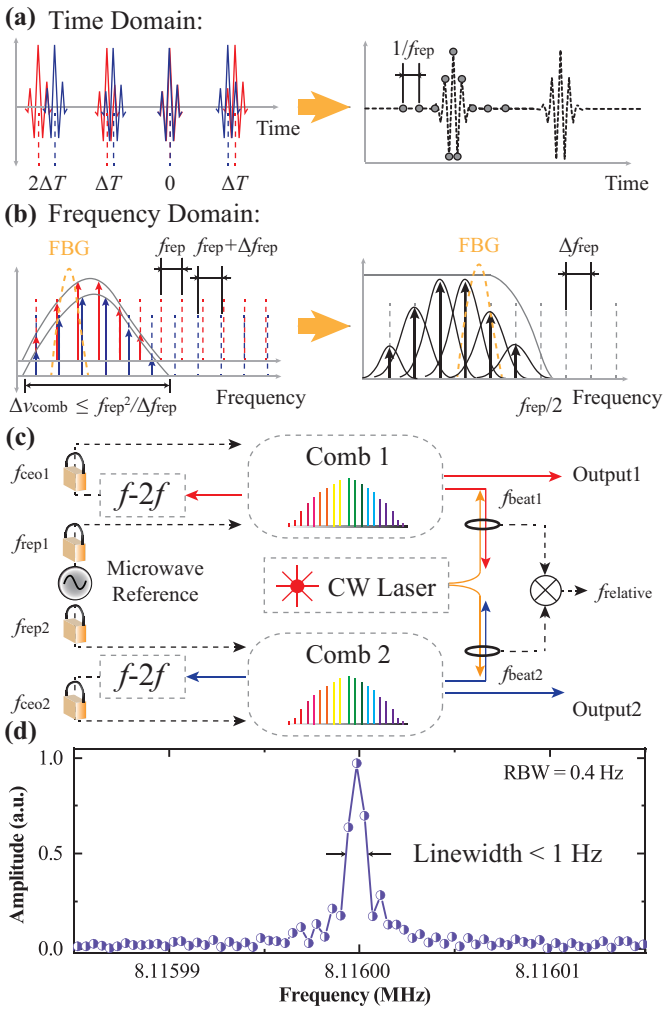



Fig. 1. The principle of dual–comb spectroscopy and digital noise compensation. (a) Time–domain picture showing the pulse–to–pulse walk–off between the two comb pulse trains to produce the interferograms periodically. (b) The equivalent frequency–domain picture with an available optical spectral bandwidth ($\Delta\nu_{\mathrm{comb}}$) and the down–converted RF comb. Solid gray line indicates filter function applied in the RF and optical to avoid aliasing effects. Solid black curves represent the relative linewidth of RF comb. (c) Schematic of dual–comb light source used in our sensing system. The relative beat signal ($f_{\mathrm{relative}}$) generated by the two combs and a CW laser is used to compensate for the mutual noise between the two combs. (d) The relative linewidth within 1 Hz of comb–resolved FBG–reflected RF spectrum in our experiment with post–correction algorithm. $f-2f$: an $f-2f$ self–reference interferometer.

comb asynchronously samples the repetitive pulse train from the other comb. Therefore, the time–domain interference between the two combs produces interferograms (IGMs) periodically, which can be detected by a PD.

Figure 1(b) provides the complementary frequency–domain picture of the DCS. The intensity and phase of the detected RF comb teeth are proportional to the product of the electric fields of the two optical comb teeth. DCS effectively maps the optical spectrum of width $\Delta\nu$ to an RF spectrum of width $\Delta\nu(\Delta f_{\mathrm{rep}}/f_{\mathrm{rep}})$, so that several hundred thousand comb teeth, spanning

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tens to hundreds of THz, can be mapped to a MHz RF band and digitized. From inspection of Fig. 1, it is clear that the one–to–one mapping from the optical to the RF domain is only maintained if the optical spectral bandwidth satisfies $\Delta\nu_{\mathrm{comb}} = f_{\mathrm{rep}}^2/(2\Delta f_{\mathrm{rep}})$, which is limited by the aliasing effect. For example, if we set realistic values for DCS with repetition frequencies of 56.09 MHz and a repetition frequency difference of 1 kHz as in our system, an aliasing–limited bandwidth of 1.57 THz can be obtained. To insert FBG into the optical path of one of the comb lasers, the reflected spectrum of the FBG is engraved on the RF comb (see Fig. 1(b)). Generally, an accurate dual–comb interferometer with high mutual coherence and comb–resolved RF spectrum requires tight–locking approaches or post–correction methods to address the parameter fluctuations between the two combs. Otherwise, the linewidth of the comb teeth in the RF domain is larger than the comb–tooth spacing, i.e., $\Delta f_{\mathrm{rep}}$, which prevents us to develop a comb–resolved phase–stable dual–comb interferogram [see Fig. 1(b) for the solid black curves] and the determination of the central frequency of the FBG spectrum would be inaccurate. To obtain the comb–resolved phase–stable RF spectrum, in this study, we used an effective and low–cost post–correction algorithm based on mutual fluctuation compensation of the two combs without any tight–locking. In the dual–comb source system we used, a free–running continuous–wave (CW) laser was used as an optical intermediary to obtain the relative beat signal between the two combs, as shown in Fig. 1(c). This relative beat signal corrected the noise of $\Delta f_{\mathrm{rep}}$ by reconstructing an interferogram's sampling sequence. Attributed to this post–processing method, we can correct both the carrier phase noise and timing jitter so that a comb–resolved RF spectrum with relative linewidth < 1 Hz can be achieved, as is shown in Fig. 1(d).

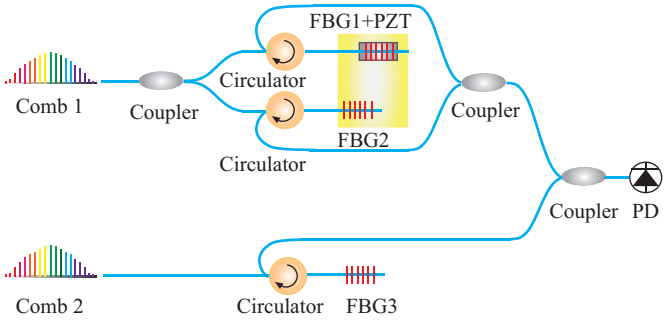



Fig. 2. Schematic diagram of the experimental setup. The Bragg wavelengths of FBG1 and FBG2 are both around 1559 nm and FWHMs are 0.07 nm. The Bragg wavelength of FBG3 is around 1559 nm and the FWHM is 1 nm. FBG1 and FBG2 are placed on a temperature adjusting device represented by the yellow block to experience the same temperature. Coupler: 50/50 fiber coupler; PZT: piezoelectric transducer; PD: photo detector.

The experimental setup for pure static strain sensing is shown in Fig. 2. Two erbium–doped mode–locked fiber ring lasers with slightly different repetition rates ($\Delta f_{\mathrm{rep}}$ = 1 kHz) were used for pure static strain sensing of the FBG sensors using DCS. As Fig. 2 shows, a pulse train from Comb 1 ($f_{\mathrm{rep1}}$ = 56.090 MHz) passed through a Michelson–based fiber interferometer and then interfered with Comb 2 ($f_{\mathrm{rep2}}$ = 56.091 MHz), generating IGMs with a certain update time $T_{\mathrm{update}} = 1/\Delta f_{\mathrm{rep}}$. In the interferometer, a 50/50 fiber coupler was used to divide the pulse train into two arms. The FBG1 with a Bragg wavelength of around 1559 nm and full–width at half maximum (FWHM) of 0.07 nm constituted the measurement arm in which a piezoelectric transducer (PZT) was used to apply static strain; the FBG2 with the same Bragg wavelength and FWHM as FBG1 acted as the reference arm for temperature compensation. The two FBGs (FBG1 and FBG2) were both placed on a temperature adjusting device. It is essential in this method to place the two FBGs close to each other, so that they experience the same temperature.

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Due to the optical path difference (OPD) between the two arms (see Fig. 2), the pulse trains from reference arm and measurement arm are separated in the time domain. Then, these two pulse trains were guided into another 50/50 fiber coupler via optical circulators and sampled by the local oscillation (Comb 2), generating both the reference IGM sequence and measurement IGM sequence with a certain update time of ~1ms, as the repetition frequency difference was 1 kHz. Therefore, in a time window of $1/\Delta f_{\mathrm{rep}}$, the obtained reference IGM and measurement IGM are separated in the time domain with the time delay determined by the OPD between the two arms. Here, the FBG3 (Bragg wavelength = 1559 nm, FWHM = 1 nm) was used for optical spectrum filtering to improve the signal–to–noise ratio (SNR) of the IGMs.

### 2.2. The principle of pure static sensing

FBGs are obtained by creating periodic variations in the refractive index of the core of an optical fiber. When light passes through the grating at the Bragg wavelength, the light reflected by the varying zones of the refractive index will be in phase and amplified. When strain is induced in an FBG, the relative change in the Bragg wavelength is expressed as:

$$\frac{\Delta\lambda_{\mathrm{B}}}{\lambda_{\mathrm{B}}} = (1-\rho_{\mathrm{e}})\varepsilon, \tag{1}$$

where $\varepsilon$ is the longitudinal strain on the FBG and $\rho_{\mathrm{e}}$ is the effective photo–elastic constant of the fiber core material. The Bragg wavelength $\lambda_{\mathrm{B}}$ is also affected by temperature changes. The relative change in the Bragg wavelength due to a temperature change is expressed as:

$$\frac{\Delta\lambda_{\mathrm{B}}}{\lambda_{\mathrm{B}}} = (\alpha+\xi)\Delta T = k\times\Delta T, \tag{2}$$

where $\Delta T$ is the change in temperature experienced at the FBG location, $\alpha$ is the thermal expansion, $\xi$ is the thermo–optic coefficient, and $k$ is the effective temperature–dependent coefficient. By combining Eqs. (1) and (2), we obtain the effective Bragg wavelength shift due to both strain and temperature as:

$$\frac{\Delta\lambda_{\mathrm{B}}}{\lambda_{\mathrm{B}}} = k\times\Delta T + (1-\rho_{\mathrm{e}})\varepsilon. \tag{3}$$

For pure strain measurements, the effect of the temperature change on the Bragg wavelength has to be compensated for. In this study, FBG2 is used to detect the temperature–induced wavelength shift, which is expressed as $\Delta\lambda_{\mathrm{B2}}/\lambda_{\mathrm{B2}} = k_2\times\Delta T$. To use the Bragg wavelength shift for temperature compensation, the determination of the temperature response relationship between the two FBGs is the key–point. When no strain is applied to FBG1, the wavelength shift of FBG1 can also be expressed as $\Delta\lambda_{\mathrm{B1}}/\lambda_{\mathrm{B1}} = k_1\times\Delta T$. Then, we can obtain the temperature response ratio of the two FBGs as:

$$\frac{\Delta\lambda_{\mathrm{B1}}/\lambda_{\mathrm{B1}}}{\Delta\lambda_{\mathrm{B2}}/\lambda_{\mathrm{B2}}} = \frac{k_1\times\Delta T}{k_2\times\Delta T} = K_{\mathrm{T}}. \tag{4}$$

Therefore, the strain– and temperature–induced wavelength shift of FBG1 can be expressed as:

$$\frac{\Delta\lambda_{\mathrm{B1}}}{\lambda_{\mathrm{B1}}} \approx \frac{\Delta\nu_{\mathrm{B1}}}{\nu_{\mathrm{B1}}} = K_{\mathrm{T}}\times\frac{\Delta\nu_{\mathrm{B2}}}{\nu_{\mathrm{B2}}} + (1-\rho_{\mathrm{e}})\varepsilon, \tag{5}$$

where $\nu_{\mathrm{B1}}$ corresponds to the Bragg optical frequency of FBG1. As in the dual–comb interferometer, we perform a Fourier transform of every obtained IGM after coherent averaging and determine the central frequency of the reflected RF spectrum ($f_{\mathrm{RF}}$) by finding the mode frequency nearest to the spectral peak. The change in the Bragg optical frequency can be expressed by the

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change in the Bragg radio frequency as $\Delta\nu_B = \Delta f_{RF}/\Delta f_{rep} \times f_{rep1}$. Therefore, the relative optical frequency change can be expressed as:

$$\frac{\Delta\lambda_B}{\lambda_B} \approx \frac{\Delta\nu_B}{\nu_B} = \frac{\Delta f_{RF}}{\nu_B}\frac{f_{rep1}}{\Delta f_{rep}} \propto \frac{\Delta f_{RF}}{\nu_B}, \tag{6}$$

where the symbol "$\propto$" indicates the direct proportionality. As a result, the frequency shift of FBG1 induced by pure strain is then denoted as:

$$\Delta f_{strain} = \Delta f_1 - K_T \times \Delta f_2. \tag{7}$$

Here, the frequency shift $\Delta f$ can be either the optical frequency or RF. In our system, the change in the radio central frequency for 1 kHz corresponds to 56.09 MHz in the optical frequency domain. For convenience, we give the experimental results in the RF domain in the following sections.

## 3. Results

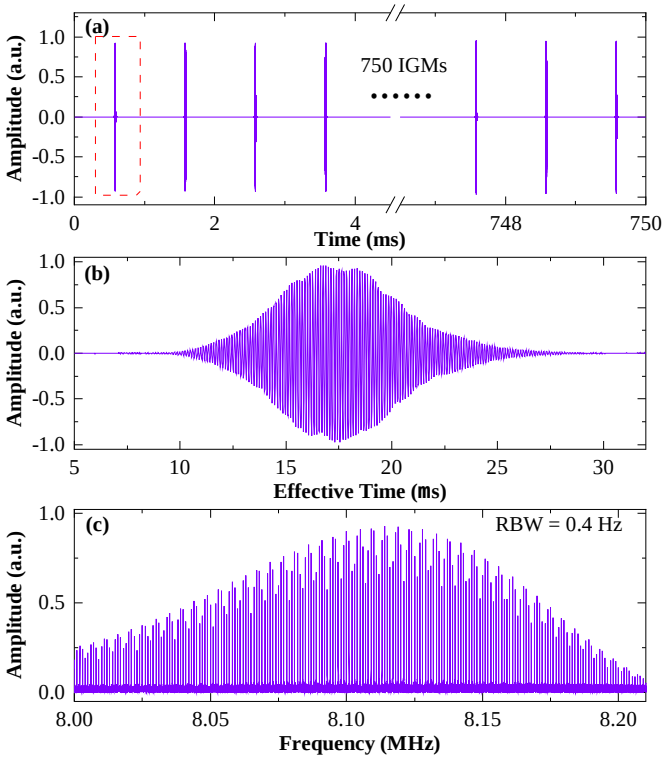



Fig. 3. Time–domain and frequency–domain interference results of the measurement arm. (a) The interferograms (IGMs) obtained by measuring the FBG1–reflected pulses and local oscillator pulses interference. The coherent–averaging time is 0.35 s. (b) The enlarged view of the first IGM surrounded by a red dashed box in (a). (c) The corresponding RF spectrum of the IGMs in (a). A comb–resolved phase–stable RF spectrum is obtained via a fast Fourier transform. Note that the periodic fringes in the spectrum are caused by the multiple reflection pulses in the fiber path.

By digitizing the signal from the PD, the IGM sequence with an update time of ~1ms is acquired. Due to the optical path difference between the reference arm and the measurement

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arm, the interference signals of the two arms are separated in the time domain. Therefore, we can distinguish and extract the IGMs from any of the two arms. Initially, we show the results of the measurement arm in both the time domain and frequency domain. Figure 3(a) shows the interference signals after 0.35 s coherent averaging. Figure 3(b) shows an enlarged view of the first IGM surrounded by a red dashed box in Fig. 3(a). The RF spectrum is obtained via a fast Fourier transform (see Fig. 3(c)) of totally 750 IGMs in Fig. 3(a). Benefitting from the post–correction method, we can get the comb–resolved phase–stable RF spectrum. Then, the central frequency of FBG can be determined between adjacent comb teeth.

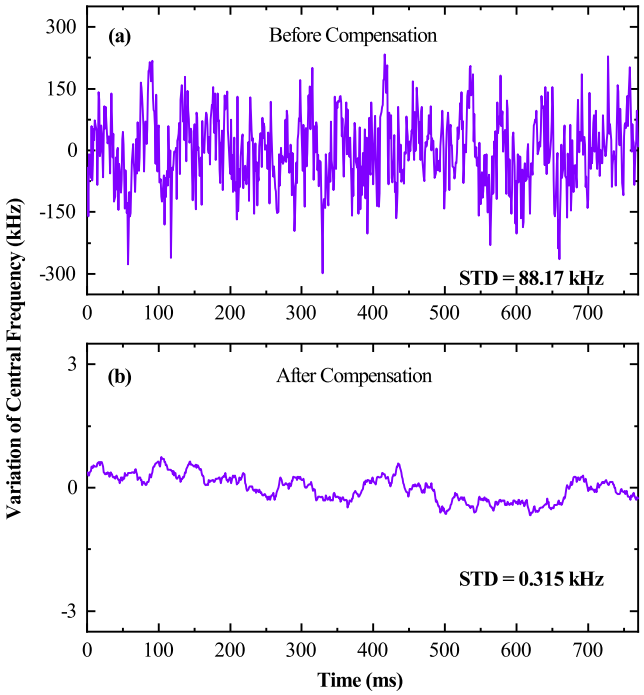



Fig. 4. Comparison of the variation of the Bragg radio–frequency of the FBG with no strain applied. (a) Before digital noise compensation, the variation range is within ±300 kHz, and the stability is 88.17 kHz. (b) After digital noise compensation, the variation range is within ±1 kHz with the stability of 0.315 kHz.

Figure 4 shows the comparison of the central frequency of one FBG before and after correcting for the phase noise and timing jitter of the IGMs. Here, no strain was applied to the measurement arm, i.e., FBG1. Nevertheless, there was no active control of the temperature or mechanical vibration in our experiment. The IGM sequence was coherently averaged by 350 ms and then a Fourier transform was performed to obtain the data points in Fig. 4. The difference between the two curves in Fig. 4 is the result of random environmental changes and measurement errors due to relative frequency noise of DCS. Here, we used the standard deviation (STD) to represent the stability of the measured radio Bragg frequency. Before digital compensation (see Fig. 4(a)), the central frequency of the reflected RF spectrum of FBG1 fluctuates within ±300 kHz and has an STD of 88.17 kHz; after noise compensation, the fluctuation range of the central frequency of FBG1 is reduced to ±1 kHz with an STD of 0.315 kHz (see Fig. 4(b)). That is, in the optical frequency domain, the Bragg frequency of FBG1 oscillates within a mode interval without an applied strain, which demonstrates that the spectral resolution of this strain sensing system can reach the comb repetition frequency of ~56 MHz.

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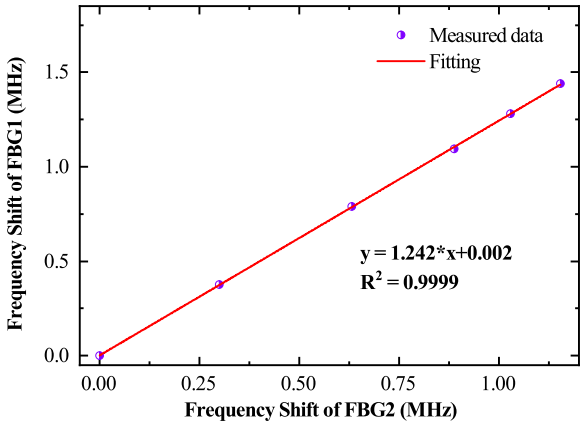



Fig. 5. The determination of the ratio of the temperature response ($K_T$) of the two FBGs with a temperature range around 43 °C. The correlation coefficient (0.9999) shows good linearity of the response, and the slope (1.242) refers to the ratio of the temperature response of the two FBGs.

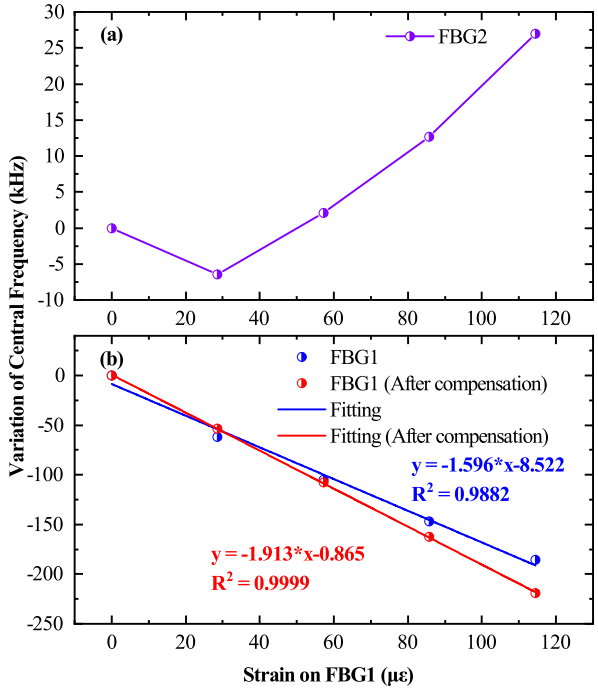



Fig. 6. The performance of temperature compensation of our system. (a) Temperature–induced frequency shift of the monitor FBG (FBG2) versus the strain value applied on FBG1. (b) Performance of temperature compensation of sensing FBG (FBG1). The correlation coefficient of fitting is optimized from 0.9882 to 0.9999.

To use FBG2 as a temperature monitor, it is essential to determine the ratio of the temperature–induced Bragg frequency shift between FBG1 and FBG2 because manufacturing error may exist. In this experiment, the thermoelectric cooler (TEC) powered by a voltage source was used to

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change the environment temperature of the two FBGs with a range around 43 °C. Figure 5 shows the temperature–induced central frequency shift of the reflected RF spectra of the two FBGs. The semi–solid violet dots represent the measured data and the solid red line represents the linear fitting. The slope is 1.242 with a correlation coefficient ($R^2$) of 0.9999. That is to say, the temperature response ratio $K_T$ in Eq. (7) is 1.242.

To evaluate the performance of temperature compensation of our system, a temperature drift was induced to change the central frequency of the reference FBG by 33 kHz. Figure 6(a) depicts the variation of Bragg RF of the temperature effect of FBG2. Figure 6(b) shows the measured data and linear fitting results before and after temperature compensation with a strain step of 30 με until 120 με. After temperature compensation of the measurement arm, the correlation coefficient ($R^2$) of fitting is optimized from 0.9882 to 0.9999.

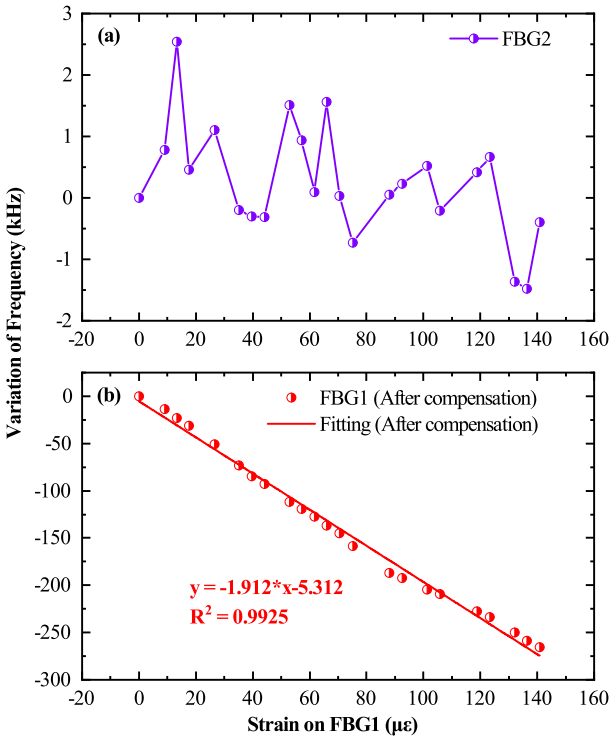



Fig. 7. Characteristics of the pure strain sensor of our system. (a) Temperature–induced frequency variation of FBG2 versus the strain value applied on FBG1. (b) Bragg radio–frequency shift of the sensing FBG after temperature compensated. The strain sensing sensitivity (slope of fitting) is 1.912 kHz/με, corresponding to 0.85 pm/με of spectral sensitivity.

Then, we adjusted the voltage of the PZT to apply strain to FBG1 with a range of 140 με to evaluate the characteristics of our system as a pure strain sensor. In our system, the measurement range of the applied strain is limited by the output range of the PZT controller, not by the FBG. In Fig. 7, the temperature–induced frequency shift of FBG2 is about 5 kHz. Meanwhile, the variation range of the central frequency of FBG1 is approximately 265 kHz after compensating for the temperature effect. As is shown, with a smaller strain step, the frequency shifts of FBG1 still exhibit a good linear relationship with the applied strain in the measurement range of 140 με. The slope coefficient of fitting is 1.912 kHz/με in the RF domain, corresponding to 0.11

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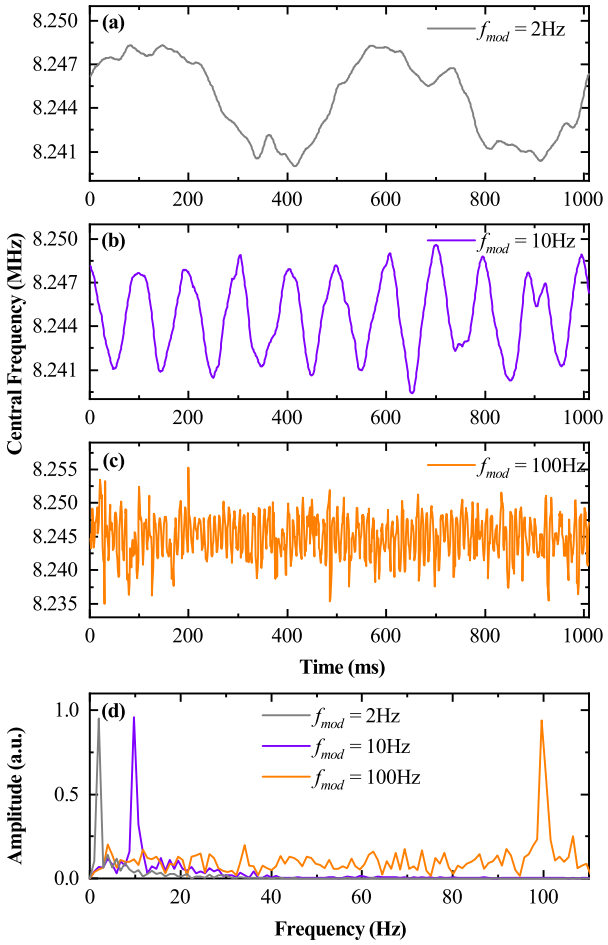



Fig. 8. Evaluation of the response rate of the strain sensor in our system. (a) Time–domain modulation response of the Bragg RF at $f_{\mathrm{mod}}$ of 2 Hz. (b) Time–domain modulation response of the Bragg RF at $f_{\mathrm{mod}}$ of 10 Hz. (c) Time–domain modulation response of the Bragg RF at $f_{\mathrm{mod}}$ of 100 Hz. The vertical axes of (a) – (c) denote the Bragg radio–frequency of FBG1, and the horizontal axes represent the lab time. (d) Response rate spectra via fast Fourier transform of (a) – (c). The modulation–frequency peaks at 2Hz, 10 Hz and 100 Hz can be observed.

GHz/με in the optical frequency and 0.85 pm/με of spectral sensitivity. The good accordance of high sensitivity with Fig. 6(b) also demonstrates the validity of our method. The reason for the deviation from linear slope in Fig. 7(b) is because the display resolution of PZT–applied strain is low (about 0.16 με). Here, the strain value (horizontal axes in Fig. 7(b)) was converted from the voltage value that was read directly from the screen of the voltage controller. And the linearity of the PZT drive is also imperfect. This problem can be solved by using another strain applier, e.g. a precise displacement stage. Since the stability of the central radio frequency is better than 1 kHz, the measurement stability of strain sensing should exceed 0.5 με theoretically.

After evaluating the sensing stability and sensitivity of our system, we estimated the response rate and strain resolution of the sensor. By applying a voltage to the piezoelectric actuator with

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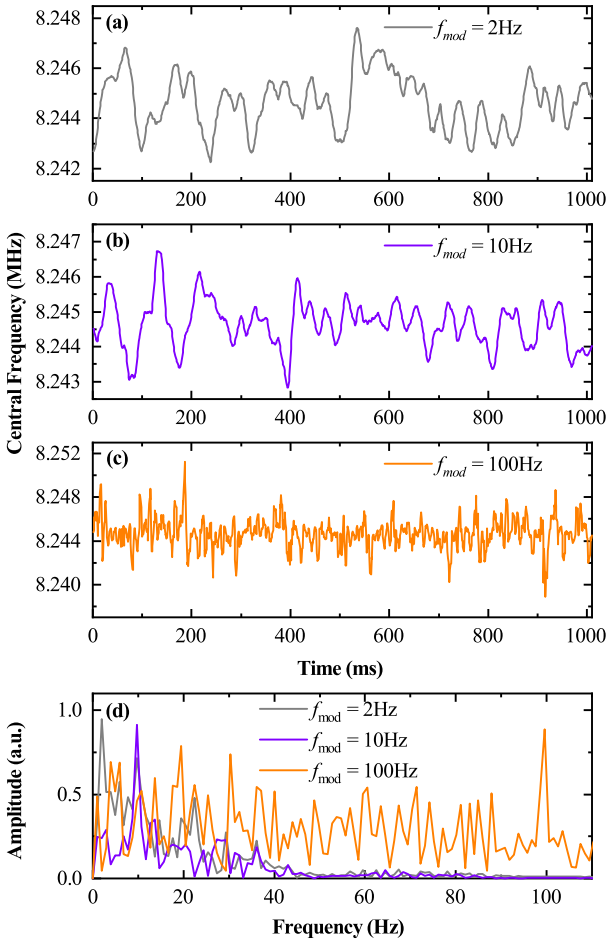



Fig. 9. Evaluation of the strain resolution of the strain sensor in our system. (a) Time–domain response of the Bragg RF at $f_{\mathrm{mod}}$ of 2 Hz. (b) Time–domain response of the Bragg RF at $f_{\mathrm{mod}}$ of 10 Hz. (c) Time–domain response of the Bragg RF at $f_{\mathrm{mod}}$ of 100 Hz. The vertical axes of (a) – (c) denote the Bragg radio frequency of FBG1, and the horizontal axes represent the lab time. (d) Response rate spectra via fast Fourier transform of (a) – (c). The modulation–frequency peaks at 2 Hz, 10 Hz and 100 Hz can be observed, which illustrates the static and dynamic strain resolution are both better than 0.8 με experimentally.

certain modulation frequencies ($f_{\mathrm{mod}}$) of the FBG1, we caused a known deformation (5 με) in the sensor FBG to assess its response rate. Since the update rate of the IGM signals is 1 kHz ($\Delta f_{\mathrm{rep}}$), the number of IGMs for coherent averaging was set as $\Delta f_{\mathrm{rep}}/5f_{\mathrm{mod}}$. Figures 8(a) – 8(c) show the measurements performed in the time domain for excitation of 5 με at $f_{\mathrm{mod}}$ values of 2 Hz, 10 Hz, and 100 Hz respectively; a cosine modulation of the demodulated Bragg radio–frequency was observed. The time–domain central frequency variations are also proportional to the deformation. The modulation spectra are shown in Fig. 8(d) via fast Fourier transform of Figs. 8(a) – 8(c). The distinct modulation–frequency peaks at 2 Hz, 10 Hz and 100 Hz well illustrate the modulation response characteristic of the proposed system. To experimentally estimate the strain resolution of the sensing system, we then reduced the strain modulation amplitude to 0.8 με and the modulation

frequencies were still 2 Hz, 10 Hz, and 100 Hz, respectively. The modulation frequency peaks can be distinguished with a relatively low SNR in Fig. 9(d), whereas the time–domain central Bragg radio–frequency variations are difficult to recognize in Figs. 9(a) – 9(c). The spectra in Fig. 9(d) have a lower SNR than the spectra of 5 με modulation (Fig. 8(d)) due to the higher proportion of noise such as detector noise and intensity noise. Here, we have demonstrated that both the static (< 10 Hz) and dynamic strain resolution are better than 0.8 με experimentally. Generally, the dynamic resolution is supposed to be better than the static resolution because the oscillation frequency is far from $1/f$ noise. However, sensing resolutions at higher modulation frequencies than 100 Hz are difficult to be verified experimentally under the current parameter setting of repetition frequency difference of 1 kHz. For modulation frequencies higher than 100 Hz, the number of measured points in one cycle is less than 10, which sets up difficulties for averaging time for a high SNR. Fortunately, our results are sufficient to estimate the static and dynamic resolution of the system.

## 4. Discussion

In this section, we will discuss the factors that influence the stability of the determination of the Bragg frequency of FBGs and the strain sensing resolution. The experiment results indicated that the strain resolution was better than 0.8 με. It is well known the spectral stability and resolution have a large influence on the measurement error of the central Bragg frequency and the sensing resolution. The simulation was performed to analyze the effect of the SNR and bandwidth of the interference signal. The results are shown in Fig. 10. First, IGMs with a certain carrier–wave frequency $f_c$ and different bandwidths (FWHM) were generated. By applying different noise level, we changed the SNR of the IGMs. Here, the SNR is defined as the ratio of the signal amplitude to the noise amplitude and FWHM actually refers to the bandwidth of the FBG in wavelength. Subsequently, fast Fourier transform and Gaussian fitting were applied to calculate the central Bragg frequency ($f_{cal}$). The stability (the vertical axis in Fig. 10) describes the stability of $f_{cal} - f_c$. The simulation results guide the optimization direction to obtain better frequency stability as well as strain resolution, i.e., a higher SNR or narrower bandwidth.

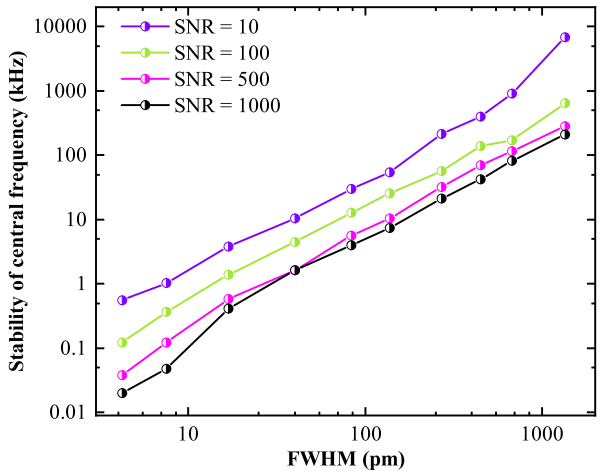



Fig. 10. The stability of the Bragg radio–frequency is simulated under different SNR and FWHM. The simulation illustrates the restrictive relation between the SNR and FWHM of the IGMs. From inspection of the simulation, the stability of Bragg RF can reach ~10 Hz under an SNR of 1000 and an FWHM of 1 pm.

The SNR can be expressed as $SNR_{NEP} \propto \sqrt{P_1 P_2}\sqrt{T}/(M \cdot NEP)$, where $NEP$ denotes the

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effective power of the detector noise, $P_1$ and $P_2$ represent the optical power of the two combs, and $M$ and $T$ denote the tooth number and averaging time respectively [14]. This expression assumes that the optical power of each comb is evenly distributed to every comb mode; therefore, the expression includes the ratio $M^{-1/2}$. In addition, the Fourier coefficient of the detector produces another ratio of $M^{-1/2}$, which results in the final factor being $M^{-1}$ [30]. If we want to improve the SNR of the signals, the tooth number $M$ and detector noise $NEP$ need to be reduced, otherwise, the optical power $P_1$ and $P_2$ or the averaging time $T$ have to be increased. For a given spectral bandwidth, a higher SNR of the IGM is preferred to obtain a more stable central Bragg frequency. When the parameters of $NEP$, $P_1$, and $P_2$ cannot be changed, a longer averaging time $T$ is usually used to optimize the results. For a fixed SNR of the IGM, the bandwidth of the sensing system is the main limitation. In our system, the FWHM of the sensing FBG we used was 0.07 nm, the reflectivity was 10%, and the SNR of the IGMs was over 200 after coherent averaging. As a result, the stability of the central frequency was less than 1 kHz, which is in good agreement with our simulation results. If we reduce the bandwidth of the FBG to 1 pm and increase the reflectivity to 70% as Kuse did [7], the SNR can be improved to over 1000; therefore, the stability of the central frequency can reach ~10 Hz. Consequently, a sensing measurement with a resolution of ~10 nε can be achieved.

For applications that require long–term measurements, the reflection spectrum of FBG3 will drift, which affects the central frequencies of the interference signals obtained. Then the demodulation results of Bragg frequencies of FBG1 and FBG2 will be influenced. Such influence has the same effect on the demodulation of the central frequencies of FBG1 and FBG2. In our experiment, the ratio of the temperature–induced Bragg frequency shift between FBG1 and FBG2 ($K_{\mathrm{T}}$) was determined. When considering the effect of the spectral drift of FBG3 on the determination, the Bragg frequency shift of FBG1 can be expressed as $\Delta\nu_{\mathrm{B1}}/\nu_{\mathrm{B1}} = k_1 \times \Delta T + \delta + (1-\rho_{\mathrm{e}})\varepsilon$, where $\delta$ is the demodulation error of the central frequency of FBG1 (or FBG2) due to the spectral drift of FBG3. When we bring the change in central frequency of FBG2 ($\Delta\nu_{\mathrm{B2}}/\nu_{\mathrm{B2}} = k_2 \times \Delta T + \delta$) into Eq. (5), the compensation result should be $K_T \times \Delta\nu_{\mathrm{B2}}/\nu_{\mathrm{B2}} + (1-\rho_{\mathrm{e}})\varepsilon = k_1 \times \Delta T + K_{\mathrm{T}} \times \delta + (1-\rho_{\mathrm{e}})\varepsilon$. As $K_{\mathrm{T}}$ is close to 1 in our system, the variation of central frequency in reference arm (FBG2) can not only compensate the change in center frequency of FBG1 caused by temperature change, but also compensate most of the effect of the spectral drift of FBG3. In practical applications, we can choose the FBGs with a closer temperature–dependent coefficient, i.e., $K_{\mathrm{T}} = k_1/k_2 = 1$, as the reference arm and measurement arm to achieve more precise compensation.

When compared to usual FBG strain sensing with an ASE source or a tunable continuous–wave laser demodulated by an OSA, our DCS–based FBG strain sensing system has the advantages of sensing sensitivity, acquisition speed, and resolution. In our system, we achieved the spectral sensitivity of the sub–picometer per micro–strain, which is higher than the typical strain sensitivity of 1.2 pm/με of usual FBG sensing systems [9]. Besides, the interference signal for determining the central frequency of the reflection spectrum can be obtained every time window of $1/\Delta f_{\mathrm{rep}}$. Normally, thousands of interference signals can be acquired in 1 second. However, the scanning of the OSA and tuning of the continuous–wave laser need over 1 s for one measurement. In our experiments, the static strain resolution is better than 0.8 με. Moreover, a sensing measurement with a resolution of ~10 nε can be achieved theoretically according to the simulation results. Nonetheless, strain sensing for slow–changing phenomena can be very challenging for usual FBG sensing systems, as it must contend with laser frequency instabilities and drifts that degrade the signal–to–noise ratio. Otherwise, attempts toward laser noise suppression have to be made for quasi–static sensing [8].

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## 5. Conclusion

In this paper, we have demonstrated an FBG–based pure strain sensing system based on DCS with digital mutual noise correction. DCS with a broad bandwidth of 1.57 THz was used for the determination of strain and temperature–induced spectral shifts of the FBGs. With this system, the stability of the measurement of the Bragg frequency shift improved significantly to 0.315 kHz, which is smaller than $\Delta f_{rep}$, thereby ensuring that the central frequency of the FBG–reflected spectrum can be distinguished within the adjacent comb teeth. High strain sensitivity of 1.912 kHz/με and static strain sensing resolution better than 0.8 με were achieved experimentally with a measurement range of 140 με which is limited by the voltage supplier. Factors that influence the frequency stability and the strain resolution have been discussed at length; a strain sensing resolution of ~nε can be achieved by using a narrower bandwidth of the FBG or a higher SNR of the IGMs. The use of DCS, which serves as both the light source and the demodulation tool, results in a compact and high–performance sensing system. This system also has the potential to provide simultaneous measurements of strain and temperature after accurate calibration of the temperature. In the future, the problem of the limited measurement range will be solved by using another strain applier and a strain sensing network with distributed FBGs will be constructed using DCS with digital noise correction.

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